SEMESTER 3 • PROGRAM CORE • BILINGUAL HANDBOOK

Electronic Circuits

A complete syllabus-based digital textbook for transistor biasing, single-stage and multistage amplifiers, tuned and power amplifiers, feedback, sinusoidal oscillators, transistor switching, multivibrators, Schmitt trigger and UJT relaxation oscillator—supported by Malayalam notes, formulas, diagrams and solved design problems.

COURSE CODE **3043**	COURSE TITLE **Electronic Circuits**
SEMESTER **3**	CREDITS **4**
CATEGORY **Program Core**	PERIODS / WEEK **4 (L:3, T:1, P:0)**
PERIODS / SEMESTER **60**	MAIN LEVEL **Applying**

60

Total periods

Four course outcomes, twenty-one module outcomes, fifty-eight teaching hours and two series-test periods.

COURSE PURPOSE

What this course develops

Amplifier design

Choose biasing, calculate the operating point and develop single-stage and cascaded transistor amplifiers.

Biasing □□□□□□□□□□□□□□□□, operating point □□□□□□□□□□□□□, single-stage, multistage transistor amplifiers □□□□□□□□□□ □□□□□□□□.

Frequency selection

Use resonance and tuned circuits to amplify a selected frequency band.

Resonance, tuned circuits □□□□□□□ □□□□□□□□□□□ □□□□□□□□□ frequency band □□□□□□□ amplify □□□□□□□□.

Power delivery

Compare amplifier classes and understand output power, efficiency, impedance matching and crossover distortion.

Power amplifier classes, output power, efficiency, impedance matching, crossover distortion □□□□□□□ □□□□□□□□□ □□□□□□□□.

Stable gain

Apply negative feedback to improve gain stability, bandwidth, distortion and impedances.

Negative feedback □□□□□□□□□□□ gain stability, bandwidth, distortion, input/output impedance □□□□□□□ □□□□□□□□□□□□□□□□□□.

Waveform generation

Design RC, LC and crystal oscillators and explain their frequency-determining networks.

RC, LC, crystal oscillators □□□□□□□□□□ □□□□□□□ frequency □□□□□□□□□□□□□□□□□□ network □□□□□□□□□□□□□□□□.

Pulse and switching circuits

Use BJTs and UJTs for switching, multivibrators, Schmitt triggering and relaxation oscillation.

BJT/UJT □□□□□□□□□□□□ switching, multivibrator, Schmitt trigger, relaxation oscillator circuits □□□□□□□□□□□□□□.

SYLLABUS STRUCTURE

Module and course-outcome map

The PDF allocates 58 instructional hours plus two series-test periods, giving the stated semester total of 60 periods.

CO1 / Module 1 14 hours

CO2 / Module 2 15 hours + Test I

CO3 / Module 3 15 hours

CO4 / Module 4 14 hours + Test II

CO	Outcome	Duration	Cognitive level
CO1	Develop basic single-stage and multistage amplifiers.	14 hours	Applying
CO2	Develop basic tuned amplifiers and power amplifiers.	15 hours	Applying
CO3	Develop feedback amplifiers and sinusoidal oscillators.	15 hours	Applying
CO4	Use transistors to realise pulse and switching circuits.	14 hours	Applying

Prerequisites: AC signal representation and passive components; semiconductor theory, diodes and transistors; basic Mathematics I and II principles. All four course outcomes are strongly mapped to PO1 in the supplied syllabus.

CO1 • 14 HOURS

Module 1 — Single-Stage and Multistage Amplifiers

Biasing establishes a stable Q-point; the amplifier then converts a small AC input into a larger output without unwanted distortion.

M1.01

3 hours • Understanding

Transistor biasing, DC load line and operating point

Biasing supplies the DC voltages and currents required to keep a transistor in the active region. The selected operating point or **Q-point** should permit the signal to swing in both directions without entering cutoff or saturation.

DC load line: $V_{CE} = V_{CC} - I_C R_C$ | endpoints: (I_C=0, V_{CE}=V_{CC}) and (V_{CE}=0, I_C=V_{CC}/R_C)

Temperature and transistor β variation can move the Q-point. Stabilisation reduces this movement. Fixed bias is simple but poorly stabilised; voltage-divider bias with an emitter resistor provides much better stability.

Fixed bias

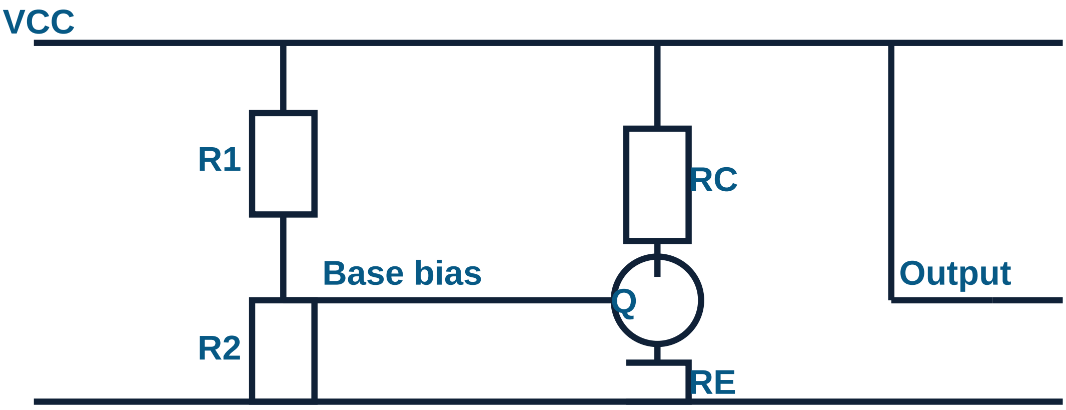
A resistor from V_{CC} to base sets base current. It uses few components but the collector current depends strongly on β and temperature.

Voltage-divider bias

R_1 and R_2 establish a nearly fixed base voltage, while R_E introduces negative DC feedback and stabilises the Q-point.

Approximate divider design: $V_B=V_{CC}R_2/(R_1+R_2)$, $V_E \approx V_B-0.7$ V, $I_E \approx V_E/R_E$, $V_{CE} \approx V_{CC}-I_C R_C-I_E R_E$

□□□□□□□ □□□□□□□□□□□□□□□□: Transistor active region-□ □□□□□□□□□□ □□□□□□□□□□□□□□□□ DC voltage/current □□□□□□□□□□□□ biasing. Q-point cutoff-□□□ saturation-□□□ □□□□□□□□□□ □□□□□□□□ output clipping □□□□□□□. Voltage-divider bias-□ emitter resistor □□□□□□□□□□ stability □□□□□□□□□□.



M1.02

4 hours • Applying

Single-stage CE amplifier with voltage-divider bias

In a common-emitter amplifier, the input is applied between base and emitter and the output is taken between collector and emitter. The stage provides voltage, current and power gain, and the output voltage is approximately 180° out of phase with the input.

$$r'_e \approx 25\ \mathrm{mV}/I_E; \quad A_v \approx -(R_C \| R_L)/r'_e; \quad A_i \approx \beta; \quad A_p = |A_v| A_i; \quad BW = f_H - f_L$$

- ✓ Input coupling capacitor blocks source DC.
- ✓ Emitter bypass capacitor increases midband voltage gain by providing an AC ground.
- ✓ Output coupling capacitor blocks collector DC from the load.
- ✓ At low frequency, coupling and bypass capacitors reduce gain.
- ✓ At high frequency, transistor and stray capacitances reduce gain.

□□□□□□□ □□□□□□□□□□□□□□□□: CE amplifier □□□□ voltage gain □□□□□□□□□; output input-□□□□ 180° phase reversal □□□□□□□□□□. Coupling capacitors DC □□□□□□. Emitter bypass capacitor AC gain □□□□□□□□□□□□□□□. Midband-□ gain □□□□□□□ constant □□□.

Solved design example — voltage-divider Q-point

Given V_{CC}=12 V, R_1=47 kΩ, R_2=10 kΩ, R_C=2.2 kΩ and R_E=1 kΩ:

1. $V_B=12\times10/(47+10)=2.105$ V.
2. $V_E\approx2.105-0.7=1.405$ V.
3. $I_E\approx1.405/1$ kΩ$=1.405$ mA and $I_C\approx1.405$ mA.
4. $V_C=12-(1.405\text{ mA}\times2.2\text{ k}\Omega)=8.909$ V.
5. $V_{CE}=8.909-1.405=7.504$ V. The Q-point is approximately (7.50 V, 1.41 mA).

Solved gain example

For I_E=1 mA and effective AC collector load 2 kΩ: $r'_e\approx25\ \Omega$, therefore $A_v\approx-2000/25=-80$. The magnitude of voltage gain is 80 and the negative sign indicates phase reversal.

M1.03

1 hour • Understanding

Emitter follower

The emitter follower or common-collector stage takes output from the emitter. Its voltage gain is slightly less than unity, but it provides high input impedance, low output impedance and large current gain.

$$A_v\approx1;\quad A_i\approx\beta+1$$

Applications: buffering, impedance matching, isolation between stages and driving a low-resistance load.

□□□□□□□: Emitter follower-□□ voltage gain □□□□□□□ 1 □□□. High input impedance, low output impedance □□□□□□□□□□ buffer, impedance matching stage □□□ □□□□□□□□□□□□□□□□.

M1.04

3 hours • Applying

Multistage amplifiers and coupling methods

When one stage cannot provide sufficient gain or impedance transformation, two or more stages are cascaded. The output of one stage drives the input of the next.

$$A_{v,total}=A_{v1}A_{v2}\ldots A_{vn};\quad \text{Gain(dB)}=20\log_{10}|A_v|;\quad \text{total dB gain = sum of stage dB gains}$$

Coupling	Strength	Limitation	Typical use
RC coupling	Good audio-frequency response, inexpensive, compact.	Poor impedance matching and reduced gain at frequency extremes.	Voltage amplifiers.
Transformer coupling	Excellent impedance matching and power transfer.	Bulky, costly and narrower frequency response.	Power-output stages.
Direct coupling	Passes DC and very low frequencies.	Drift and bias interaction between stages.	DC amplifiers, instrumentation and IC circuits.

□□□□□□□: Multistage amplifier-□ stages cascade □□□□□□□□□□□. Overall gain = □□□ stage gain-□□□□□□□□ product. RC coupling audio voltage amplification-□□□□; transformer coupling impedance matching-□□□□; direct coupling DC/low-frequency signals-□□□□ □□□□□□□□□□.

Solved overall-gain example

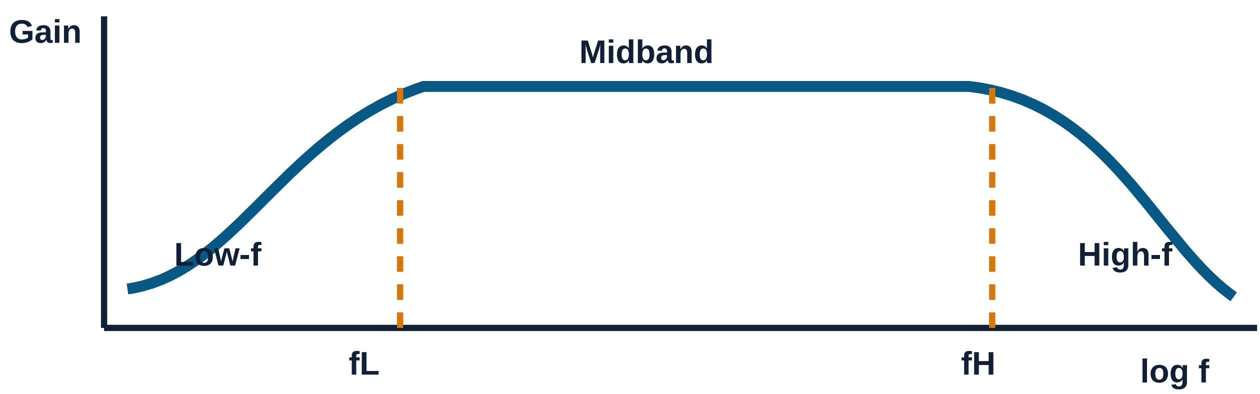
For $A_{v1}=20$ and $A_{v2}=30$, total gain=600. Gain in dB=20 $\log_{10}(600)$=55.56 dB.

M1.05–M1.06

3 hours • Understanding / Remembering

Frequency response and applications of multistage amplifiers

The response has three regions. In the low-frequency region, coupling and bypass capacitor reactance is significant. In the midband region, gain is nearly constant. At high frequency, transistor junction and stray capacitances cause gain reduction. Cascading stages generally narrows the overall bandwidth.



Applications: audio preamplifiers, microphone amplifiers, instrumentation channels, communication receivers, sensor interfaces and driver stages.

□□□□□□□: Frequency response-□ low, midband, high □□□□□ □□□□□□□ □□□□□□□ □□□□□□. Bandwidth = $f_H - f_L$. Stages □□□□□□□□□□□□□□□ overall bandwidth □□□□□□□ □□□□□□□.

Module 1 exam pattern: Draw the bias circuit, locate the Q-point, write the CE gain expressions, compare coupling schemes and solve at least one Q-point or cascaded-gain problem.

CO2 • 15 HOURS + SERIES TEST I

Module 2 — Tuned Amplifiers and Power Amplifiers

Tuned amplifiers select a narrow frequency band; power amplifiers deliver substantial AC power to a load efficiently.

M2.01–M2.02

6 hours • Applying / Understanding

Resonance, single-tuned and double-tuned amplifiers

At resonance, inductive and capacitive reactances cancel. A series resonant circuit has minimum impedance and maximum current. A parallel resonant circuit has maximum impedance and behaves as a selective load for a tuned amplifier.

$f_r=1/(2\pi\sqrt{LC})$; $Q=f_r/BW$; $BW=f_r/Q$; series $Q=\omega_0 L/R=1/(\omega_0 CR)$

Single-tuned amplifier

One parallel LC circuit forms the collector load. Gain peaks near resonance. It is simple and selective, but the gain–bandwidth compromise limits performance.

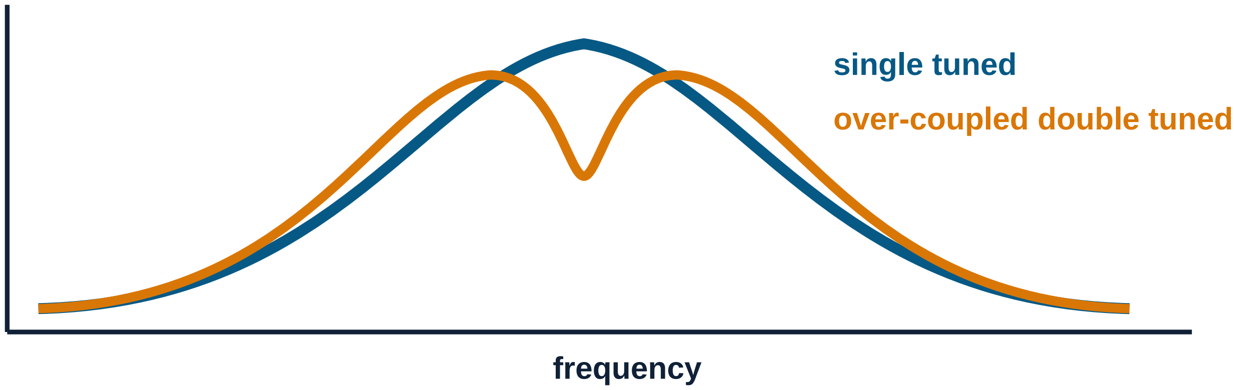
Double-tuned amplifier

Two magnetically coupled tuned circuits provide improved gain and adjustable bandwidth. Under-coupling gives one narrow peak, critical coupling gives a broad flat peak, and over-coupling gives two peaks.

□□□□□□□: Resonance-□ L, C reactance-□□□ cancel □□□□□□□□. Single tuned amplifier-□ □□□ LC tank; double tuned-□ coupled □□ □□□□□□ tanks. Q □□□□□□□□□□ selectivity □□□□□□□□□ bandwidth □□□□□□□□□□ □□□□□□□□.

Solved tuned-circuit example

For L=100 µH and C=100 pF: $f_r=1/[2\pi\sqrt{(100\times10^{-6}\times100\times10^{-12})}]=1.59$ MHz. If Q=50, BW=1.59 MHz/50=31.8 kHz.



M2.03–M2.05

9 hours • Applying / Understanding / Remembering

Power amplifiers, classes and push-pull operation

A voltage amplifier mainly raises signal voltage; a power amplifier transfers maximum useful AC power to the load. Power stages use large-signal operation, heat dissipation and impedance matching.

Class	Conduction angle	Efficiency / feature	Main use
A	360°	Best linearity; low efficiency. Maximum theoretical efficiency: 25% with resistive collector load, 50% with transformer coupling.	Low-distortion small-power stages.
B	180°	Maximum theoretical push-pull efficiency 78.5%; crossover distortion near zero crossing.	Audio power output.
AB	More than 180°, less than 360°	Reduces crossover distortion; efficiency between Class A and B.	Most practical audio output stages.
C	Less than 180°	High efficiency but severe waveform distortion; requires tuned load.	RF power amplification.

Single-ended versus double-ended

A single-ended stage uses one active device for the complete output. A double-ended or push-pull stage uses two devices on alternate half cycles. Push-pull cancels even harmonics, increases output power and avoids DC magnetisation in a centre-tapped output transformer, but requires matched devices and phase splitting.

$$\text{Class-B push-pull: } P_{o,max}=V_{CC}^2/(2R_L);\quad P_{dc}=2V_{CC}I_m/\pi;\quad \eta_{max}=\pi/4=78.5\%$$

□□□□□□□: Voltage amplifier signal voltage □□□□□□□□□□□□□□□□□; power amplifier load-□□□□□□ power □□□□□□□□□. Class A linear □□□□□ efficiency □□□□□. Class B push-pull efficiency □□□□□□□□□□□□□□□ crossover distortion □□□□□□□□□. Class AB □□□□□ □□□□□□□□□□□□□□. Class C RF tuned circuits-□ □□□□□□□□□□□□□□□.

Solved Class-B example

For V_{CC}=12 V and R_L=8 Ω, maximum AC output power is $12^2/(2\times8)=9$ W. Peak load current is 12/8=1.5 A. DC input power is $2\times12\times1.5/\pi=11.46$ W. Efficiency=9/11.46=78.5%.

Common mistake: Do not state that Class B has zero distortion. Ideal devices produce a dead zone around zero due to base-emitter threshold; Class AB bias is used to reduce this crossover distortion.

Series Test I focus: resonance and Q, single versus double tuning, power-amplifier classes, Class-B push-pull working, efficiency expressions, advantages, disadvantages and applications.

CO3 • 15 HOURS

Module 3 — Feedback Amplifiers and Sinusoidal Oscillators

Feedback controls amplifier behaviour; positive regenerative feedback is used with a frequency-selective network to sustain oscillation.

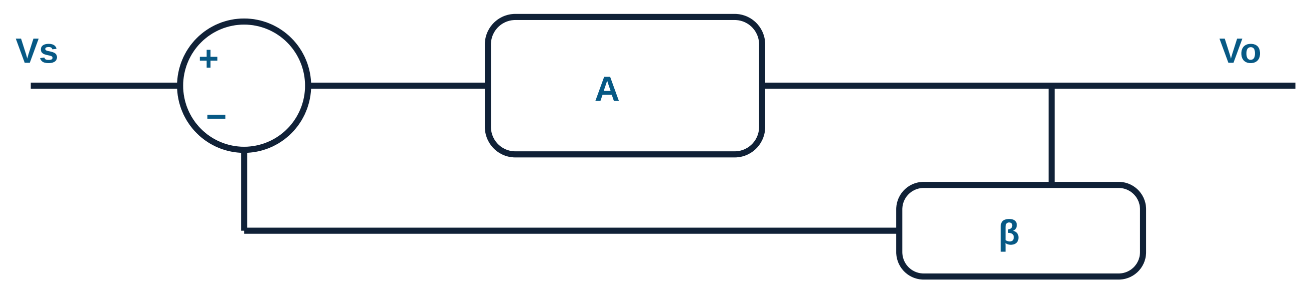
M3.01–M3.03

7 hours • Understanding

Positive and negative feedback

Feedback returns a fraction β of the output to the input. If the returned signal aids the input, feedback is positive. If it opposes the input, feedback is negative.

Negative feedback gain: $A_f = A/(1+A\beta)$ | Positive feedback gain: $A_f = A/(1-A\beta)$



Benefits of negative feedback

- ✓ Stabilises closed-loop gain against device and temperature variation.
- ✓ Reduces nonlinear distortion and internally generated noise approximately by the desensitivity factor 1+Aβ.
- ✓ Increases bandwidth.
- ✓ Allows input and output impedance to be modified by the connection topology.

Topology	Input mixing	Output sampling	Z_{in}	Z_{out}
Voltage-series	Series	Voltage	Increases	Decreases
Voltage-shunt	Shunt	Voltage	Decreases	Decreases
Current-series	Series	Current	Increases	Increases
Current-shunt	Shunt	Current	Decreases	Increases

The syllabus specifically includes typical current-series and voltage-series transistor feedback circuits.

□□□□□□: Output-□□□□ □□□ □□□□ input-□□□□□□ □□□□□□ □□□□□□□□□□□ feedback. Negative feedback gain □□□□□□□□□□□□□□□□□ stability, bandwidth □□□□□□ □□□□□□□□□□□□□□□□□ distortion □□□□□□□□□□□□□ □□□□□□□. Series mixing input impedance □□□□□□□; shunt mixing □□□□□□□□.

Solved feedback example

Open-loop gain A=100 and β=0.04. Loop gain Aβ=4. Closed-loop gain A_f=100/(1+4)=20. The desensitivity factor is 5, so parameter variation and distortion are reduced by approximately five times.

M3.04–M3.05 8 hours • Applying / Remembering

Oscillation principle and Barkhausen criterion

An oscillator converts DC supply power into an AC waveform without an external periodic input. A frequency-selective network returns a correctly phased signal to the amplifier.

Barkhausen criterion: |Aβ|=1 and total loop phase shift = 0° or 360° at the oscillation frequency.

Oscillator	Frequency expression	Key point	Application
RC phase-shift	$f=1/(2\pi RC\sqrt{6})$ for three equal RC sections	Amplifier gives 180° and RC network gives another 180°.	Audio-frequency sine generation.
Wien bridge	$f=1/(2\pi RC)$ for equal R and C	Low distortion; amplifier gain condition is approximately 3.	Audio test oscillator.
Hartley	$f=1/[2\pi\sqrt{C(L_1+L_2+2M)}]$	Inductive divider supplies feedback.	RF signal generation.
Colpitts	$f=1/(2\pi\sqrt{LC_{eq}})$, $C_{eq}=C_1C_2/(C_1+C_2)$	Capacitive divider supplies feedback.	Stable RF oscillator.
Crystal	Set mainly by crystal mechanical resonance	Piezoelectric crystal has extremely high Q and excellent stability.	Clocks, communication transmitters and frequency standards.

□□□□□□: Oscillator external AC input □□□□□□□ DC-□□□ □□□□□□ periodic AC waveform □□□□□□□□□□□□□□. Barkhausen condition: loop gain magnitude 1, phase shift 0°/360°. RC oscillators audio frequency-□□□□, LC oscillators RF-□□□□, crystal oscillator □□□□ stable frequency-□□□□ □□□□□□□□□□□□□□□.

Design example — RC phase-shift oscillator at 1 kHz

Choose C=0.01 μF=10 nF. R=1/(2πfC√6)=1/[2π×1000×10 nF×√6]=6.50 kΩ. Use a nearby standard value such as 6.49 kΩ or 6.8 kΩ and verify the practical frequency.

Design example — Wien bridge oscillator at 1 kHz

For C=0.01 μF, R=1/(2πfC)=15.9 kΩ. Use R≈15.9 kΩ in both arms and set the amplifier closed-loop gain near 3.

Applications of sinusoidal oscillators

Signal generators, radio transmitters and receivers, local oscillators, clocks, function generators, audio testing, carrier generation and measurement instruments.

Module 3 exam pattern: Derive A_f, explain the four feedback topologies, list negative-feedback effects, state Barkhausen criterion and design at least one RC or LC oscillator.

CO4 • 14 HOURS + SERIES TEST II

Module 4 — Pulse and Switching Circuits

Transistors can be driven between cutoff and saturation to act as electronic switches and to generate rectangular, pulse and relaxation waveforms.

M4.01

1 hour • Understanding

Transistor as a switch

State	Condition	Equivalent
Cutoff	Base current approximately zero; $I_C \approx 0$; $V_{CE} \approx V_{CC}$.	Open switch
Saturation	Sufficient base drive; $V_{CE(sat)}$ small, typically about 0.2 V.	Closed switch

Forced base design: $I_B \geq I_C/\beta_{forced}$; $R_B = (V_{drive} - V_{BE})/I_B$

□□□□□□□: Cutoff-□ transistor OFF/open switch; saturation-□ ON/closed switch. Reliable saturation □□□□□□ □□□□□□□□□ base current □□□□□□.

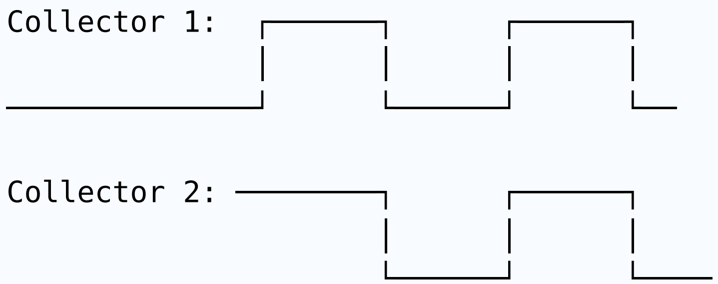
M4.02

4 hours • Applying

BJT astable multivibrator

An astable multivibrator has no stable state. Two transistor stages switch alternately through cross-coupled capacitors, producing complementary rectangular collector waveforms.

$$T = 0.693(R_1C_1 + R_2C_2)\text{; symmetrical: } T \approx 1.386RC \text{ and } f \approx 0.721/(RC)$$



Applications: clock generator, LED flasher, pulse generator, square-wave source and timing circuits.

□□□□□□□: Astable multivibrator-□□ stable state □□□□. □□□□□ transistors □□□□□□□□ ON/OFF □□□□□□□□□□□ continuous square wave □□□□□□□□.

Design example — 1 kHz symmetrical astable

Choose C=0.01 µF. R=0.721/(fC)=0.721/(1000×10 nF)=72.1 kΩ. Select 72 kΩ. Each half-cycle is approximately 0.693RC≈0.499 ms, giving T≈0.998 ms.

M4.03

5 hours • Understanding

Monostable and bistable multivibrators

Monostable

One stable state and one temporary quasi-stable state. A trigger causes one output pulse; the circuit automatically returns after the timing interval.

Typical pulse width ≈0.69RC, depending on circuit form.

Applications: one-shot pulse, time delay, pulse stretching and missing-pulse detection.

Bistable

Two stable states. A trigger changes the stored state and another trigger resets it. It acts as a one-bit memory element.

Applications: flip-flop, counter, switch debouncing and digital storage.

□□□□□□□: Monostable-□□ □□□ stable state; trigger □□□□□□□□□□□□□ □□□ pulse □□□□□□□□□□□ □□□□□□□□□□□□□□□. Bistable-□□ □□□□□□ stable states; □□□ bit memory □□□□□ □□□□□□□□□□□□□□□□□□□□.

M4.04

2 hours • Applying

Schmitt trigger

A Schmitt trigger is a regenerative comparator with two switching thresholds. It changes state at the upper trigger point (UTP) for a rising input and at the lower trigger point (LTP) for a falling input.

Hysteresis voltage $V_H = V_{UTP} - V_{LTP}$

Hysteresis prevents repeated switching due to noise near a single threshold. Applications include waveform shaping, noisy-signal restoration, level detection and switch debouncing.

□□□□□□□: Schmitt trigger-□□ UTP, LTP □□□□ □□□□□ threshold □□□□□. □□□□□□ □□□□□□□□□□□□□ hysteresis voltage. Noise □□□□ slow input-□□ clean square wave □□□□□□ □□□□□□□□□□□□□□□.

Simple threshold example

If UTP=3 V and LTP=1 V, hysteresis voltage=3−1=2 V.

M4.05

2 hours • Applying

UJT relaxation oscillator

A capacitor charges through R toward V_{BB}. When its voltage reaches the UJT peak voltage, the UJT turns on and rapidly discharges the capacitor. When current falls below the valley current, the UJT switches off and charging starts again. Capacitor voltage is sawtooth-like; the output pulse at a base terminal is sharp.

$$V_P \approx \eta V_{BB} + V_D; \quad T \approx RC \ln[1/(1-\eta)] \text{ (neglecting diode term)}; \quad f \approx 1/\{RC \ln[1/(1-\eta)]\}$$

□□□□□□□: Capacitor R □□□ charge □□□□□□□. Voltage UJT peak voltage □□□□□□□□□□ UJT ON □□□ capacitor □□□□□□□□ discharge □□□□□□□. □□□□□□□ sawtooth waveform, sharp pulse □□□□□□ □□□□□□□□.

Design example — approximately 1 kHz

Let η=0.6 and C=0.1 μF. $R=1/[fC \ln(1/(1-\eta))]=1/[1000\times0.1\ \mu F\times\ln(2.5)]=10.9\ k\Omega$. Use about 11 kΩ and verify against actual UJT parameters.

Applications: ramp generation, triggering SCRs/triacs, timing and simple sweep circuits.

Series Test II focus: cutoff and saturation, astable design, monostable/bistable operation, Schmitt-trigger thresholds and UJT relaxation-oscillator frequency.

FORMULA BANK

Essential equations and design sequence

Bias and CE amplifier

$V_B = V_{CC} R_2 / (R_1 + R_2)$

$V_E \approx V_B - 0.7$

$I_E \approx V_E / R_E$

$r'_e \approx 25\text{ mV} / I_E$

$A_v \approx -R'_C / r'_e$

Multistage and bandwidth

$A_{total} = A_1 A_2 \ldots$

$G_{v,dB} = 20 \log |A_v|$

$BW = f_H - f_L$

Tuned circuit

$f_r = 1 / (2\pi\sqrt{LC})$

$Q = f_r / BW$

$BW = f_r / Q$

Feedback

$A_f = A / (1 + A\beta)$

$\text{Desensitivity} = 1 + A\beta$

$A_{f,+} = A / (1 - A\beta)$

Oscillators

RC phase shift: $f=1/(2\pi RC\sqrt{6})$
Wien: $f=1/(2\pi RC)$
Hartley: $f=1/(2\pi\sqrt{LC_T})$
Colpitts: $C_{eq}=C_1C_2/(C_1+C_2)$

Pulse circuits

Astable: $T=0.693(R_1C_1+R_2C_2)$
Schmitt: $V_H=UTP-LTP$
UJT: $f\approx 1/[RC\ \ln(1/(1-\eta))]$

Design habit: write the target frequency or gain, choose one practical component value, calculate the remaining value, select a standard value, and finally verify the achieved result and component ratings.

FINAL REVISION

One-page memory map

Module 1

- Bias and Q-point
- Load line
- CE parameters
- Emitter follower
- RC / transformer / direct coupling
- Frequency response

Module 2

- Series and parallel resonance
- f_r, Q and BW

- Single / double tuned
- Classes A, B, AB, C
- Push-pull and efficiency
- Crossover distortion

Module 3

- $A_f = A/(1+A\beta)$
- Four feedback topologies
- Negative-feedback benefits
- Barkhausen criterion
- RC, Wien, Hartley, Colpitts
- Crystal oscillator

Module 4

- Cutoff and saturation
- Astable design
- Monostable / bistable
- UTP, LTP and hysteresis
- UJT charging/discharging
- Pulse applications

Malayalam last-day cue: Bias → gain → coupling → resonance → power classes → feedback → Barkhausen → oscillators → switch → multivibrators → Schmitt → UJT □□□□ □□□□□□□□□□ □□□□□□□□□□□□□.

Likely 1-mark areas

- Define Q-point.
- State emitter-follower voltage gain.
- Write resonant-frequency formula.
- State maximum Class-B efficiency.
- State Barkhausen criterion.
- Define hysteresis voltage.

Likely short answers

- Fixed versus divider bias.
- RC versus transformer coupling.
- Single versus double tuned.
- Class A/B/AB/C comparison.
- Negative-feedback advantages.
- Monostable versus bistable.

Likely design questions

- CE Q-point and voltage gain.
- Overall multistage gain in dB.
- LC resonance and bandwidth.
- RC phase-shift / Wien oscillator.
- Astable multivibrator.
- UJT relaxation oscillator.

PRACTICE MODEL QUESTION PAPER

ELECTRONIC CIRCUITS • Maximum Marks: 75 • Time: 3 Hours

This is a fresh practice model prepared from the supplied syllabus and is not an official board question paper.

PART A — Answer all questions. Each carries 1 mark. [5 × 1 = 5]

1. What is the purpose of transistor biasing?
2. Write the relation between resonant frequency, Q and bandwidth.
3. State the maximum theoretical efficiency of a Class-B push-pull amplifier.
4. State the Barkhausen criterion.
5. Define hysteresis voltage of a Schmitt trigger.

PART B — Answer any five. Each carries 8 marks. [5 × 8 = 40]

1. Explain voltage-divider bias and calculate the Q-point for V_{CC}=12 V, R_1=47 kΩ, R_2=10 kΩ, R_C=2.2 kΩ and R_E=1 kΩ.

2. Explain the CE amplifier, its gains, impedances and frequency response.
3. Compare RC, transformer and direct coupling. Two stages have gains 20 and 30; find total gain and dB gain.
4. Explain single- and double-tuned amplifiers. A circuit has L=100 μH, C=100 pF and Q=50; calculate f_r and bandwidth.
5. Compare Classes A, B, AB and C and explain crossover distortion.
6. Explain negative feedback and the four feedback topologies. Find A_f for A=100 and β=0.04.
7. Explain astable, monostable and bistable multivibrators with applications.

PART C — Answer any two. Each carries 15 marks. [2 × 15 = 30]

1. Explain transistor biasing, single-stage CE amplification and multistage coupling. Include design equations and frequency response.
2. Explain RC phase-shift, Wien bridge, Hartley, Colpitts and crystal oscillators. Design an RC phase-shift oscillator for 1 kHz using C=0.01 μF.
3. Explain transistor switching, BJT astable multivibrator, Schmitt trigger and UJT relaxation oscillator. Design a 1 kHz symmetrical astable using C=0.01 μF.

COMPLETE ANSWER KEY

Model-paper answers

Part A answers

No.	Answer
1	To establish and stabilise a suitable DC operating point in the active region.
2	$Q=f_r/BW$, therefore $BW=f_r/Q$.
3	78.5%.
4	Loop-gain magnitude $\lvert A\beta\rvert=1$ and total loop phase shift 0° or 360°.
5	$V_H=V_{UTP}-V_{LTP}$.

Part B full answers

B1. Voltage-divider bias and Q-point

Voltage-divider bias uses R1 and R2 to establish a nearly fixed base voltage. RE introduces negative DC feedback and improves thermal and beta stability.

$V_B = V_{CC}\, R_2/(R_1+R_2)$
$= 12\times 10/(47+10)$
$= 2.105$ V

$V_E = V_B - 0.7 = 1.405$ V
$I_E = V_E/R_E = 1.405$ mA
$I_C \approx I_E = 1.405$ mA

$V_C = V_{CC} - I_C R_C$
$= 12 - (1.405\text{ mA} \times 2.2\text{ k}\Omega)$
$= 8.909$ V

$V_{CE} = V_C - V_E = 8.909 - 1.405 = 7.504$ V

Q-point ≈ (V_{CE}=7.50 V, I_C=1.41 mA).

The emitter resistor opposes a temperature-induced rise of current: increased current raises VE, reduces VBE and limits the original rise.

B2. CE amplifier

1. Input is applied to base–emitter and output is taken from collector–emitter.
2. Voltage-divider resistors establish the Q-point.
3. Coupling capacitors pass AC but block DC.
4. The emitter bypass capacitor increases midband gain.
5. Output is approximately 180° out of phase with input.

Important relations:
$r_e' \approx 25\ mV/I_E$
$A_v \approx -(R_C||R_L)/r_e'$
$A_i \approx \beta$
$A_p = |A_v|A_i$
$Z_{in} \approx R_1||R_2||r_\pi$
$Z_{out} \approx R_C$
$BW = f_H - f_L$

Frequency response:
• Low frequency: capacitor reactance reduces gain.
• Midband: gain is approximately constant.
• High frequency: transistor and stray capacitances reduce gain.

B3. Coupling methods and overall gain

RC coupling:
• Good audio response, economical and compact.
• Poor impedance matching and reduced gain at extremes.

Transformer coupling:
• Good impedance matching and power transfer.
• Bulky, costly and restricted frequency response.

Direct coupling:
• Passes DC and very low frequency.
• Susceptible to drift and bias interaction.

Overall voltage gain = $20 \times 30 = 600$.
Gain in dB = $20 \log_{10}(600) = 55.56$ dB.

B4. Tuned amplifiers and numerical answer

A single-tuned amplifier uses one parallel LC collector load and gives maximum gain near resonance. It is simple but has a gain–bandwidth limitation.

A double-tuned amplifier uses two coupled resonant circuits. Coupling controls the response: under-coupled gives a narrow single peak, critical coupling gives a broader useful response, and over-coupled gives two peaks.

Given L=100 μH and C=100 pF:
$f_r = 1/(2\pi\sqrt{LC})$
$= 1.59$ MHz

$Q = f_r/BW$
$BW = f_r/Q = 1.59\ \text{MHz}/50 = 31.8\ \text{kHz}$.

B5. Power-amplifier classes

Class A: conducts 360°, excellent linearity, low efficiency; maximum 25% with resistive load and 50% with transformer coupling.

Class B: conducts 180°. Two devices in push-pull reproduce the full cycle. Maximum theoretical efficiency is 78.5%.

Class AB: conducts more than 180° but less than 360°. A small standing current reduces crossover distortion.

Class C: conducts less than 180°. It is highly nonlinear but efficient and is used with an LC tuned load in RF circuits.

Crossover distortion occurs around the zero crossing when neither Class-B transistor conducts because the input has not exceeded the base-emitter threshold.

B6. Negative feedback and topologies

Negative feedback returns a fraction of output in opposition to the input.

$A_f = A/(1+A\beta)$

Benefits:
- stable gain
- lower distortion and noise
- wider bandwidth
- controllable input and output impedances

Topologies:
1. Voltage-series: Zin increases, Zout decreases.
2. Voltage-shunt: Zin decreases, Zout decreases.
3. Current-series: Zin increases, Zout increases.
4. Current-shunt: Zin decreases, Zout increases.

For $A=100$ and $\beta=0.04$:
$A\beta=4$
$A_f=100/(1+4)=20$.

B7. Multivibrators

Astable:
- no stable state
- switches continuously
- produces rectangular wave
- used in clocks and pulse generators

Monostable:
- one stable and one quasi-stable state
- trigger produces one pulse and circuit returns automatically
- used in timers, pulse stretching and delay circuits

Bistable:
- two stable states
- set/reset triggers select the state
- used as a flip-flop, one-bit memory and counter element.

Part C full answers

C1. Biasing, CE amplifier and multistage amplifier

Introduction:
An amplifier needs a stable DC Q-point and an AC signal path. Biasing fixes the DC conditions; coupling and bypass components determine AC behaviour.

A. Biasing
• Need: active-region operation and undistorted signal swing.
• DC load line: $V_{CE}=V_{CC}-I_C R_C$.
• Q-point is the no-signal operating point.
• Fixed bias is simple but beta dependent.
• Voltage-divider bias gives better stability through a nearly fixed V_B and emitter feedback.

Approximate equations:
$V_B=V_{CC}\, R_2/(R_1+R_2)$
$V_E=V_B-0.7$
$I_E=V_E/R_E$
$V_{CE}=V_{CC}-I_C R_C-I_E R_E$

B. CE amplifier
• input at base, output at collector
• high voltage and power gain
• 180° phase reversal
• coupling capacitors block DC
• emitter bypass capacitor raises AC gain

$r_e' \approx 25\text{ mV}/I_E$
$A_v \approx -(R_C||R_L)/r_e'$
$A_i \approx \beta$
$A_p=|A_v|A_i$

C. Frequency response
• low frequency: coupling/bypass effects
• midband: almost constant gain
• high frequency: junction/stray capacitance
• $BW=f_H-f_L$

D. Multistage amplifier
Need: gain greater than one stage can supply.
$A_{total}=A_1A_2\ldots$; total dB gain is the sum of stage dB gains.

Coupling:
• RC for audio voltage amplification
• transformer for impedance matching and power transfer
• direct for DC and very low frequency

Conclusion:

Correct biasing, suitable coupling and controlled frequency response are jointly required for a practical amplifier.

C2. Sinusoidal oscillators and RC design

Oscillator principle:
An oscillator uses positive feedback and a frequency-selective network to convert DC power to AC.

Barkhausen criterion:
$|A\beta|=1$ and loop phase=0°/360°.

RC phase-shift:
• transistor amplifier gives 180°
• three RC sections give another 180°
• $f=1/(2\pi RC\sqrt{6})$

Wien bridge:
• lead-lag RC bridge
• low-distortion audio output
• $f=1/(2\pi RC)$
• amplifier gain condition approximately 3

Hartley:
• inductive divider L1 and L2
• $f=1/[2\pi\sqrt{(C(L1+L2+2M))}]$

Colpitts:
• capacitive divider C1 and C2
• $Ceq=C1C2/(C1+C2)$
• $f=1/(2\pi\sqrt{LCeq})$

Crystal:
• uses piezoelectric resonance
• extremely high Q and excellent frequency stability
• used for clocks and communication frequency standards

Design for RC phase-shift oscillator:
f=1 kHz, C=0.01 μF=10 nF
$R=1/(2\pi fC\sqrt{6})$
=6.50 kΩ
Choose approximately 6.49 kΩ or a nearby practical value and verify actual frequency.

C3. Switching and pulse circuits

A. Transistor switch
• cutoff: $I_B \approx 0$, $I_C \approx 0$, open switch
• saturation: sufficient base current, $V_{CE(sat)}$ small, closed switch
• $R_B = (V_{drive} - V_{BE})/I_B$

B. Astable multivibrator
• no stable state
• cross-coupled transistor stages switch alternately
• complementary rectangular outputs
• $T = 0.693(R_1C_1 + R_2C_2)$
• symmetrical $f \approx 0.721/(RC)$

Design:
$f = 1$ kHz, $C = 0.01$ μF
$R = 0.721/(fC) = 72.1$ kΩ
Choose 72 kΩ.

C. Schmitt trigger
• regenerative comparator
• switches at UTP for rising input and LTP for falling input
• hysteresis $V_H = UTP - LTP$
• converts a noisy/slow waveform into a clean rectangular waveform

D. UJT relaxation oscillator
• capacitor charges through R
• UJT turns on at peak voltage and rapidly discharges capacitor
• UJT switches off near valley condition
• capacitor voltage is sawtooth-like
• $f \approx 1/[RC \ln(1/(1-\eta))]$

Applications:
pulse generation, timing, waveform shaping, ramp generation and SCR/triac triggering.

REFERENCES FROM THE SYLLABUS

Textbooks and online resources

Code	Book / Author
T1	R. S. Sedha, *A Text Book of Applied Electronics*, S. Chand.

Code	Book / Author
R1	N. N. Bhargava, Kulshreshtha and S. C. Gupta, *Basic Electronics and Linear Circuits*, TMH.
R2	Robert Boylestad, *Electronic Devices and Circuits*, PHI.
R3	Anil K. Maini and Varsha Agarwal, *Electronic Devices and Circuits*, Wiley India.
R4	David A. Bell, *Electronic Devices and Circuits*, PHI.
R5	Allen Mottershead, *Electronic Devices and Circuits: An Introduction*, Prentice-Hall of India.

Online resources listed in the syllabus: Electronics Tutorials, ElProCus, BrainKart Electronics Engineering and Electrical4U.